

Nour Mahmoud

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[LinkedIn](#) | [Portfolio](#)

SUMMARY

NC State Computer Science graduate (Valedictorian, 4.0 GPA) with full-stack and systems experience across React/Spring Boot web apps, a multithreaded C++ client-server engine, and C#/C++ development in Unity and Unreal Engine. Published HCI researcher seeking an entry-level software engineering role.

EDUCATION

North Carolina State University — Raleigh, NC

May 2026

Bachelor of Science in Computer Science | GPA: 4.0 / 4.0

Honors: Summa Cum Laude; Valedictorian — ranked #1 of 11,560 graduating undergraduates; #1 in Computer Science (of 678) and #1 in the College of Engineering (of 4,100)

Relevant coursework: Data Structures & Algorithms, Software Engineering, Operating Systems, Artificial Intelligence, Computer Graphics, Human-Computer Interaction

EXPERIENCE

Research Assistant, NSF REU Program (HCI)

May 2025 – May 2026

North Carolina State University — Raleigh, NC

- Annotated 2,700 dialogue turns from 10 engineering teams in Python and Jupyter to study collaboration, reaching strong inter-rater agreement (Jaccard > 0.80)
- Identified decision-making patterns linked to reduced debugging time through statistical analysis and iterative labeling
- Co-authored a research paper *accepted to the HCI 2026 Conference*

Teaching Assistant & TA Senate Member

Aug. 2023 – May 2026

North Carolina State University — Raleigh, NC

- Supported 400+ students across five semesters as a TA for Intro to Java (CSC 116) and Data Structures & Algorithms (CSC 316), debugging their code on the spot during labs and office hours
- Broke down complex topics such as recursion, sorting algorithms, and data-structure implementation into clear, accessible explanations adapted to each student's level
- Elected to the TA Senate; conducted lab observations and behavioral interviews to evaluate teaching practices and select new TAs

Computer Science Ambassador (Volunteer)

Aug. 2024 – May 2026

North Carolina State University — Raleigh, NC

- Represented the CS department at outreach events and campus tours, explaining technical topics to prospective students and families

PROJECTS

WolfCafe — Full-Stack Web Application

NC State

React | Spring Boot | REST APIs | JWT | MySQL

- Built a full-stack ordering app with a React frontend and Spring Boot backend, exposing REST APIs secured with JWT authentication and role-based access control
- Developed responsive UI components that streamlined the ordering and account-management flows
- Worked in a 6-person agile team across feature development, system design, and code review

Multiplayer Game Engine (SDL2)

NC State

C++ | ZeroMQ | Multithreading | Networking

- Designed a client-server architecture for real-time, synchronized state across multiple connected clients
- Implemented modular, reusable input, event, and update systems used across multiple games

ITSY Debugging Interface

Capstone | Katabasis (nonprofit)

Unity | C# | API Design | Software Architecture | Figma

- Designed a host-agnostic adapter API so the debugging plugin integrates with any block-based coding game, decoupling the interface from any specific host program through abstraction
- Analyzed the host system's architecture to define clean module boundaries, building reusable, independently testable components
- Implemented three learning modes in C# (code reading, step-through tracing, reflection) with quiz logic, coroutine-driven animation, and drag-and-drop categorization

Cretaceous Campaign — Paleontology Game

NC Museum of Natural Sciences

C++ | Unreal Engine 5 | Blueprints | Data-Driven Design

- Led the fossil reconstruction system: drag-and-drop bone placement, species progress tracking, and a unified inventory with duplicate detection and filtering
- Architected reusable Blueprint systems (reconstruction manager, inventory component, data-driven species registry) and a persistent HUD wired via event dispatchers for real-time updates

TECHNICAL SKILLS

Languages: Java, C#, C++, C, Python, JavaScript, SQL, HTML, CSS

Frameworks & Libraries: React, Spring Boot, Hibernate, Tomcat, Vite, JUnit, Maven, ZeroMQ

Concepts: Data Structures, Algorithms, Object-Oriented Programming, REST APIs, Client-Server Architecture, JWT Authentication, Multithreading, Networking, Unit Testing, CI/CD, Agile / Iterative Development

Tools & Environments: Git/GitHub, GitHub Actions, Jenkins, Unix/Linux, MySQL, Figma, Unity, Unreal Engine, Jupyter, Visual Studio